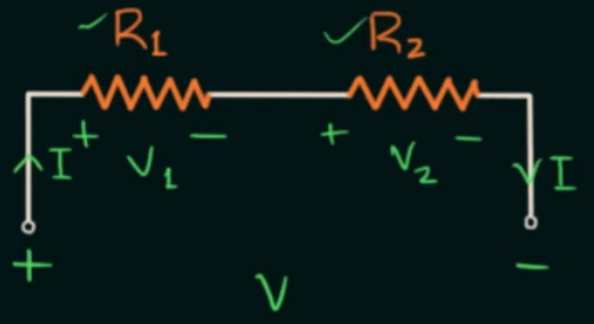
Series → I → same

Voltage Divider Rule



$$R_{eq} = R_1 + R_2$$

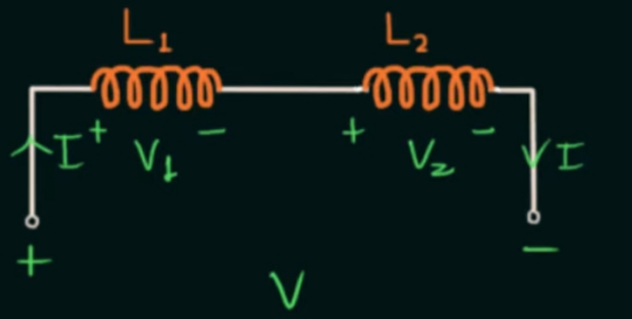
$$I = \frac{V}{R_{eq}} = \frac{V}{R_1+R_2}$$

total vm.

$$V_1 = IR_1 = \frac{V}{R_1+R_2} \cdot R_1$$ ← own res.

$$V_2 = IR_2 = \frac{V}{R_1+R_2} \cdot R_2$$

$$V_3 = \frac{V}{R_1+R_2+\dots+R_n} \cdot R_3$$

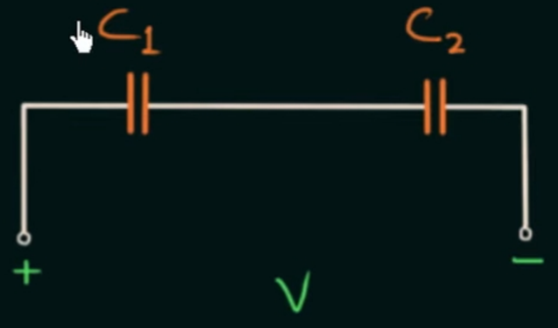


$$V = L_{eq}\frac{dI}{dt} \Rightarrow \frac{dI}{dt} = \frac{V}{L_1+L_2}$$

$$V_1 = L_1\frac{dI}{dt} = \frac{V}{L_1+L_2} \cdot L_1$$

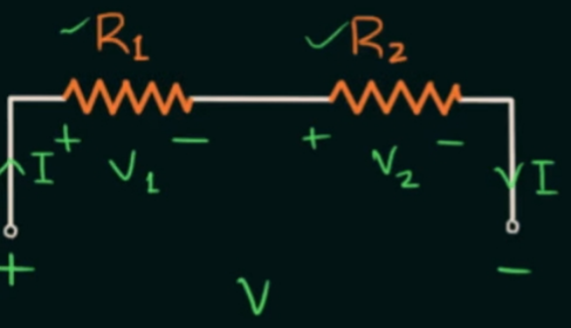
$$V_2 = L_2\frac{dI}{dt} = \frac{V}{L_1+L_2} \cdot L_2$$

$$V_5 = \frac{V}{L_1+L_2+\dots+L_n} \cdot L_5$$



Voltage Divider Rule

eries → I → same



$R_{eq} = R_1 + R_2$

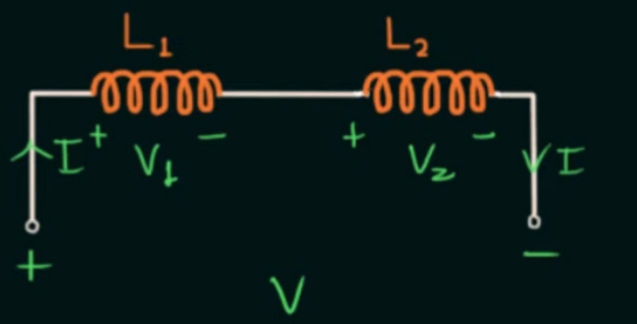
$I = \frac{V}{R_{eq}} = \frac{V}{R_1+R_2}$

total vm.

$V_1 = IR_1 = \frac{V}{R_1+R_2}\cdot R_1$ ← own res.

$V_2 = IR_2 = \frac{V}{R_1+R_2}\cdot R_2$

$V_3 = \frac{V}{R_1+R_2+\dots+R_n}\cdot R_3$

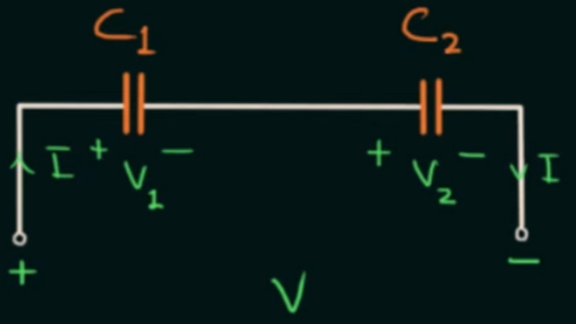


$V = L_{eq}\frac{dI}{dt} \Rightarrow \frac{dI}{dt} = \frac{V}{L_1+L_2}$

$V_1 = L_1\frac{dI}{dt} = \frac{V}{L_1+L_2}\cdot L_1$

$V_2 = L_2\frac{dI}{dt} = \frac{V}{L_1+L_2}\cdot L_2$

$V_5 = \frac{V}{L_1+L_2+\dots+L_n}\cdot L_5$



$\frac{1}{C_{eq}} = \frac{1}{C_1}+\frac{1}{C_2} \Rightarrow C_{eq} = \frac{C_1C_2}{C_1+C_2}$

$Q = C_{eq}V = \frac{C_1C_2}{C_1+C_2}\cdot V$

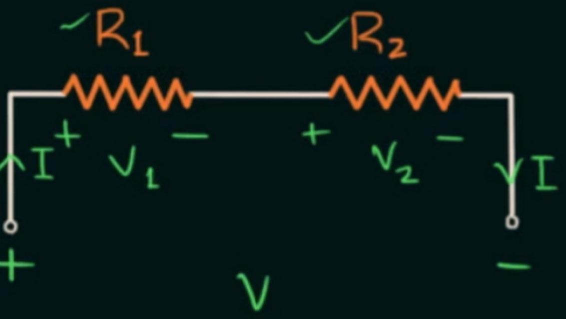
$V_1 = \frac{Q}{C_1} = \frac{C_1C_2V}{C_1+C_2}\cdot\frac{1}{C_1} = \frac{V}{C_1+C_2}\cdot C_2$

$V_2 = \frac{Q}{C_2} = \frac{V}{C_1+C_2}\cdot C_1$

H.W. $V_3 = ?$

Voltage Divider Rule

eries $\rightarrow I \rightarrow$ same



$R_{eq} = R_1 + R_2$

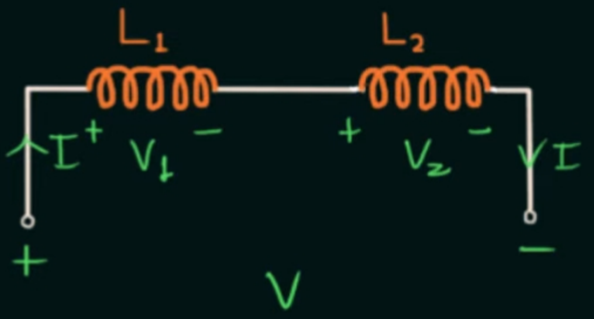
$I = \dfrac{V}{R_{eq}} = \dfrac{V}{R_1+R_2}$

total vlt.

$V_1 = IR_1 = \dfrac{V}{R_1+R_2} \cdot R_1$ own res.

$V_2 = IR_2 = \dfrac{V}{R_1+R_2} \cdot R_2$

$V_3 = \dfrac{V}{R_1+R_2+\dots+R_n} \cdot R_3$

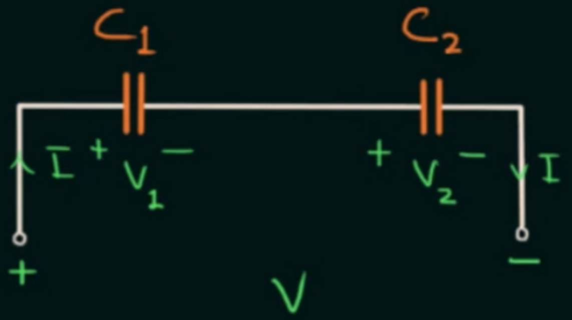


$V = L_{eq}\dfrac{dI}{dt} \Rightarrow \dfrac{dI}{dt} = \dfrac{V}{L_1+L_2}$

$V_1 = L_1\dfrac{dI}{dt} = \dfrac{V}{L_1+L_2} \cdot L_1$

$V_2 = L_2\dfrac{dI}{dt} = \dfrac{V}{L_1+L_2} \cdot L_2$

$V_5 = \dfrac{V}{L_1+L_2+\dots+L_n} \cdot L_5$



$\dfrac{1}{C_{eq}} = \dfrac{1}{C_1} + \dfrac{1}{C_2} \Rightarrow C_{eq} = \dfrac{C_1C_2}{C_1+C_2}$

$Q = C_{eq}V = \dfrac{C_1C_2}{C_1+C_2} \cdot V$

$V_1 = \dfrac{Q}{C_1} = \dfrac{C_1C_2V}{C_1+C_2} \cdot \dfrac{1}{C_1} = \dfrac{V}{C_1+C_2} \cdot C_2$

$V_2 = \dfrac{Q}{C_2} = \dfrac{V}{C_1+C_2} \cdot C_1$

H.W. $V_3 = V_{Total} \cdot \dfrac{C_{eq}}{C_3}$

Do read this for Information on How to calculate Voltage through Capacitors with more than 2 capacitors :

https://gemini.google.com/share/3ffa28f629a5